

Fibroblast-led collective invasion of carcinoma cells with differing roles for RhoGTPases in leading and following cells

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Imaging of collectively invading cocultures of carcinoma cells and stromal fibroblasts reveals that the leading cell is always a fibroblast and that carcinoma cells move within tracks in the extracellular matrix behind the fibroblast. The generation of these tracks by fibroblasts is sufficient to enable the collective invasion of the squamous cell carcinoma (SCC) cells and requires both protease- and force-mediated matrix remodelling. Force-mediated matrix remodelling depends on integrins $\alpha 3$ and $\alpha 5$, and Rho-mediated regulation of myosin light chain (MLC) activity in fibroblasts, but these factors are not required in carcinoma cells. Instead, carcinoma cells use Cdc42 and MRCK (myotonic dystrophy kinase-related CDC42-binding protein kinases) mediated regulation of MLC to follow the tracks generated by fibroblasts.

The collective invasion of carcinoma cells has been noted by pathologists for many years^{1,2}, but remains poorly understood. It has also been observed that, in many cases, invasive carcinomas retain many epithelial markers and do not upregulate mesenchymal markers; for example, collectively invading SCC cells frequently retain localization of the cell adhesion protein p120 catenin to sites of cell–cell contact³. A recent study has shown that MT1-MMP (MMP14) is required to mediate matrix proteolysis during collective invasion; however, this study used cells that were either of mesenchymal origin (HT1080) or had upregulated mesenchymal markers (MDA-MB-231)⁴. Mesenchymal cells (such as fibroblasts) are capable of matrix remodelling, and stromal fibroblasts within tumours can promote tumour progression. Here, we use an ‘organotypic’ culture model to bring these observations together. SCC cells that retain epithelial markers are unable to remodel the surrounding matrix, but instead follow behind stromal fibroblasts that remodel the extracellular matrix (ECM). We further determine differing requirements for integrins and RhoGTPases in leading and following cells.

RESULTS

Stromal fibroblasts lead collective cancer cell invasion

To study the invasion of SCC cells, a culture system that closely mimics the physiological situation was used^{2,5}. SCC cells are grown on top of a dense matrix predominantly composed of fibrillar collagen I, with lesser amounts

of laminins and collagen IV. These cultures are supported by a metal bridge to ensure that the upper surface is exposed to air and CO₂, whereas the lower surface is in contact with media (see Supplementary Information, Fig. S1a). When SCC cells are cultured in these conditions they show aberrant differentiation, but remain on top of the matrix (Fig. 1a). Introduction of fibroblasts isolated from either oral or vulval squamous cell carcinomas (HN-CAF or V-CAF, respectively) into the matrix caused two different SCC cell lines (SCC12 and A431), but not untransformed keratinocytes (HaCAT), to invade (Fig. 1a and see Supplementary Information, Fig. S1b, c). Analysis of phalloidin-stained samples demonstrated clearly that SCC cells always invaded as collective chains and retained expression of epithelial markers, such as E-cadherin and p120 catenin³ (see Supplementary Information, Fig. S1d and Movie 1). The extent of invasion was dependent on the number of fibroblasts introduced into the matrix (see Supplementary Information, Fig. S1b). Cancer cells with mesenchymal characteristics (HT1080 fibrosarcoma cells) invaded equally well in the absence or presence of fibroblasts, indicating that stromal fibroblasts are particularly crucial for the invasion of carcinoma cells that retain epithelial characteristics (see Supplementary Information, Fig. S1c). To investigate whether fibroblasts were increasing the motility of SCC cells and thereby promote invasion, multi-photon confocal time-lapse imaging of the ‘organotypic’ cultures was performed. Interestingly, SCC cells were motile, even in the absence of fibroblasts, and their motility was not significantly changed in the presence of fibroblasts

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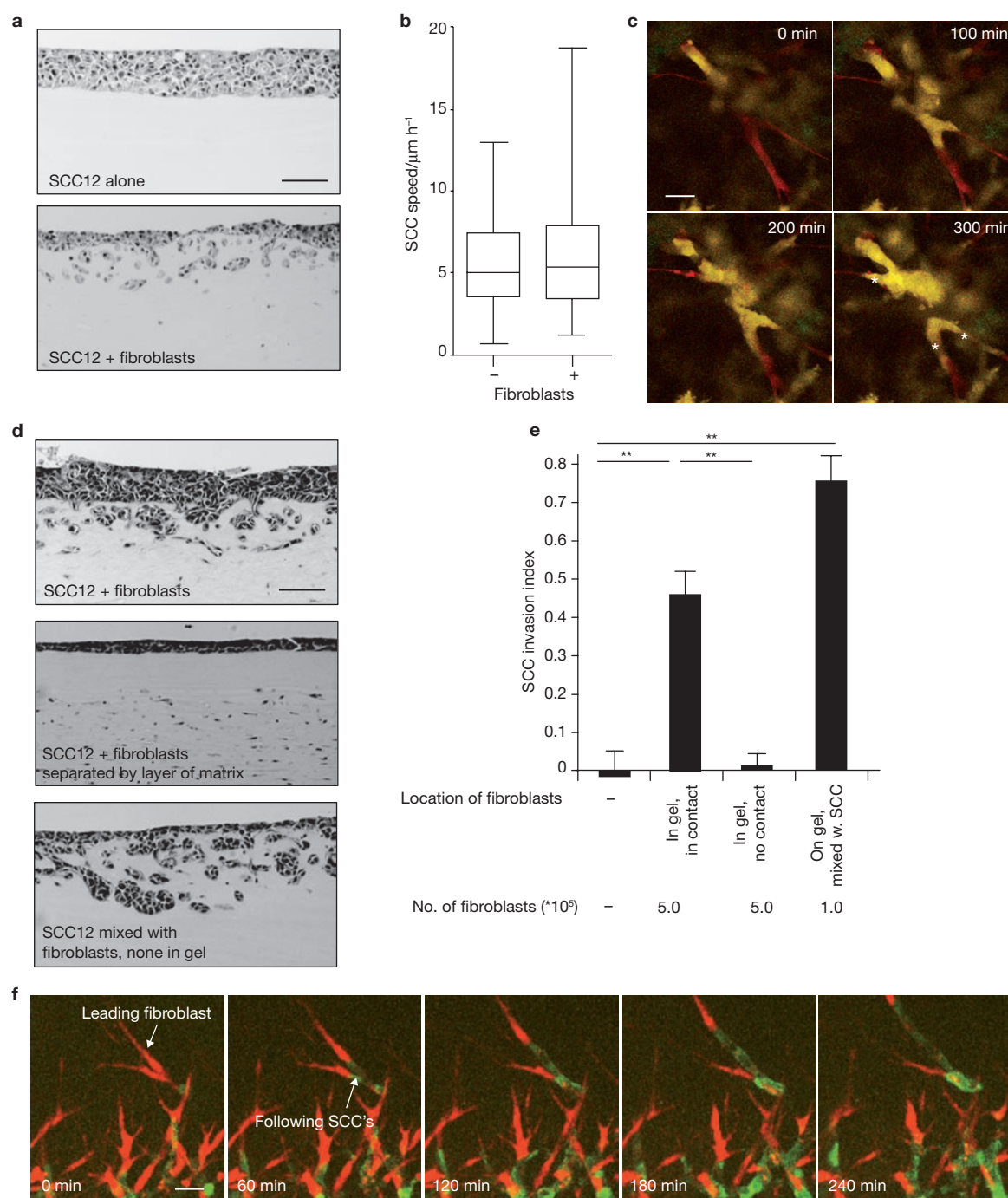


Figure 1 Fibroblasts promote and lead collective SCC invasion. **(a)** H&E-stained sections of SCC12 cells cultured in an organotypic system in the absence or presence of stromal fibroblasts. **(b)** Quantification of SCC12 cell motility in an organotypic system in the absence or presence of stromal fibroblasts. Whiskers mark 10th and 90th percentile, 'boxes' mark 25th and 75th percentile, and 80 cells were quantified for each condition. **(c)** Representative images of SCC cells (A431 cells, yellow) moving (note cells marked with an asterisk) in a matrix containing stromal fibroblasts (red). **(d)** H&E-stained sections of SCC12 cells cultured in an organotypic system in the absence of stromal fibroblasts, in the presence

of stromal fibroblasts but separated by a thin layer of matrix lacking fibroblasts, or mixed directly with stromal fibroblasts. **(e)** Quantification of SCC12-cells invasion in an organotypic system in the absence of stromal fibroblasts, in the presence of stromal fibroblasts, in the presence of stromal fibroblasts but separated by a thin layer of matrix lacking fibroblasts, or mixed directly with stromal fibroblasts ($n = 3$). The single asterisk indicates $P < 0.05$ and the double asterisk indicates $P < 0.01$. The error bars represent s.d. **(f)** Representative images of SCC12 cells (green) and stromal fibroblasts (red) invading into naïve matrix. The scale bars represent 100 μm in **a** and **d**, 20 μm in **c** and 40 μm in **f**.

(Fig. 1b). Close examination of these movies suggested that SCC cells moved in groups and were always close to fibroblasts, appearing to move 'along' them (Fig. 1c and see Supplementary Information, Movie 2).

To determine whether close proximity between SCC cells and fibroblasts was required for the invasion of SCC cells, fibroblasts were added into the matrix as before, but an additional thin layer

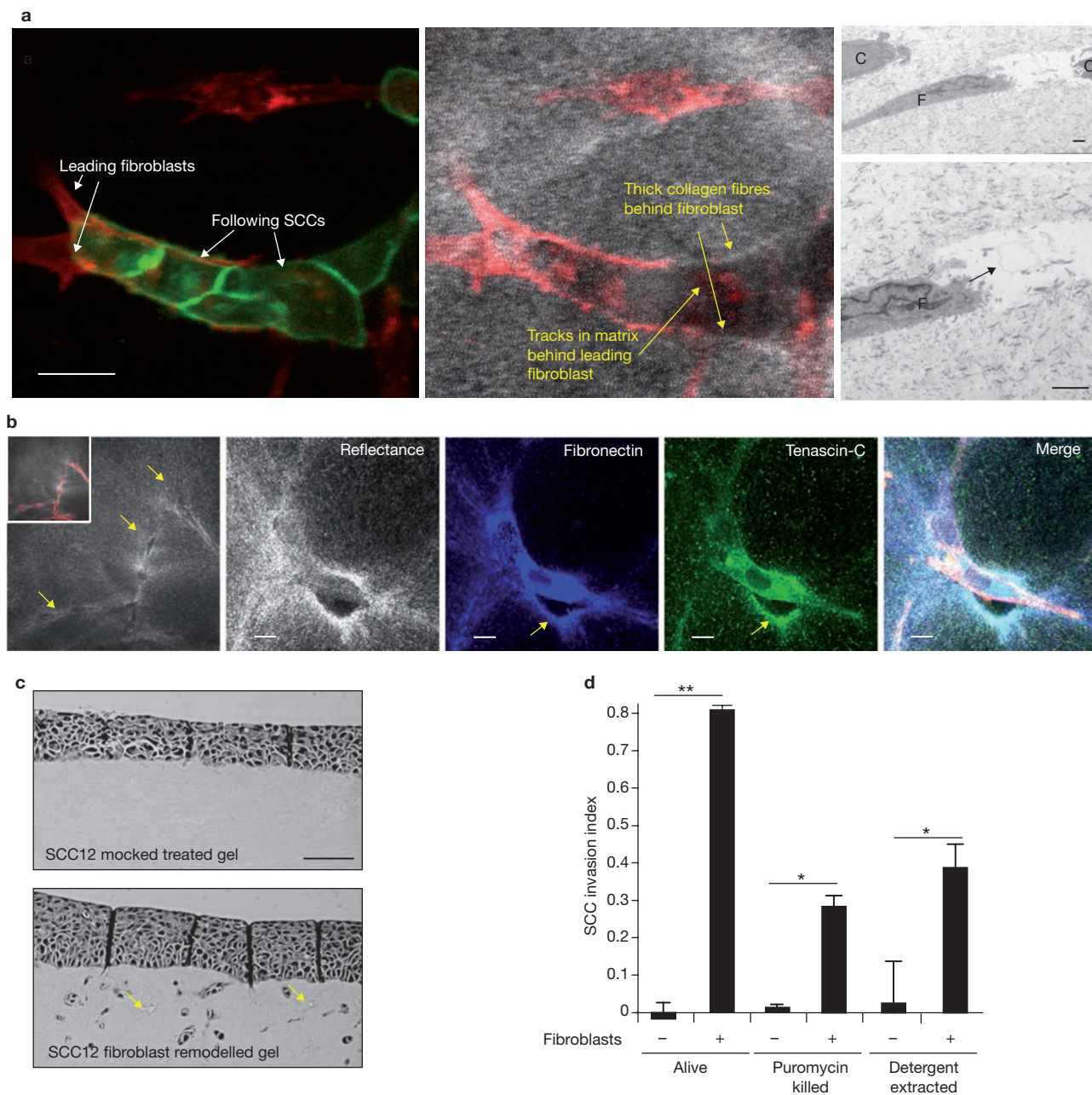


Figure 2 Fibroblasts generate tracks in the matrix that are sufficient to support SCC invasion. **(a)** Representative images of SCC12 cells transfected with GFP-CAAX (green) and HN-CAFs transfected with mRFP-CAAX (red) invading into naïve matrix (shown in white by reflectance imaging). The scale bars represent 20 μm . The right-hand panels show transmission electron microscopy of matrix, fibroblasts (F) and SCC cells (C). The arrow points to hole in matrix. The scale bar represents 2 μm . **(b)** Left-hand panel shows reflectance image of matrix after invasion of stromal fibroblasts with the inset indicating F-actin staining of fibroblasts (red). Reflectance (white), immunostaining for fibronectin (blue), immunostaining for tenascin (green)

and a merged image including F-actin (red) are also shown. Arrows indicate the tracks in the matrix and associated deposition of fibronectin and tenascin. The scale bars represent 10 μm . **(c)** H&E sections of SCC12 cells cultured on matrix that previously contained fibroblasts, which were then killed with puromycin, or that never contained fibroblasts. The arrows indicate the tracks in the matrix. The scale bar represents 100 μm . **(d)** Quantification of SCC12 invasion into matrix that previously contained fibroblasts, which were then killed with puromycin or detergent extraction, or that never contained fibroblasts ($n = 3$). The error bars represent s.d. The single asterisk indicates $P < 0.05$ and the double asterisk indicates $P < 0.01$.

(~80 μm) of matrix lacking fibroblasts was also added. This completely abolished the ability of the fibroblasts to promote the invasion of the SCC cells, suggesting that a diffusible factor produced by the fibroblasts was unlikely to be responsible for the increased SCC cell invasion (Fig. 1d, e); however, a role for factors that can only diffuse short distances (<80 μm) cannot be excluded. In agreement with this

result, fibroblast-conditioned media or a variety of growth factors were unable to promote the invasion of SCC cells (see Supplementary Information, Fig. S1e and data not shown). If close proximity between fibroblasts and SCC cells is critical for increased invasion⁶, we reasoned that directly mixing SCC cells and fibroblasts on top of the matrix should be more effective than distributing them throughout

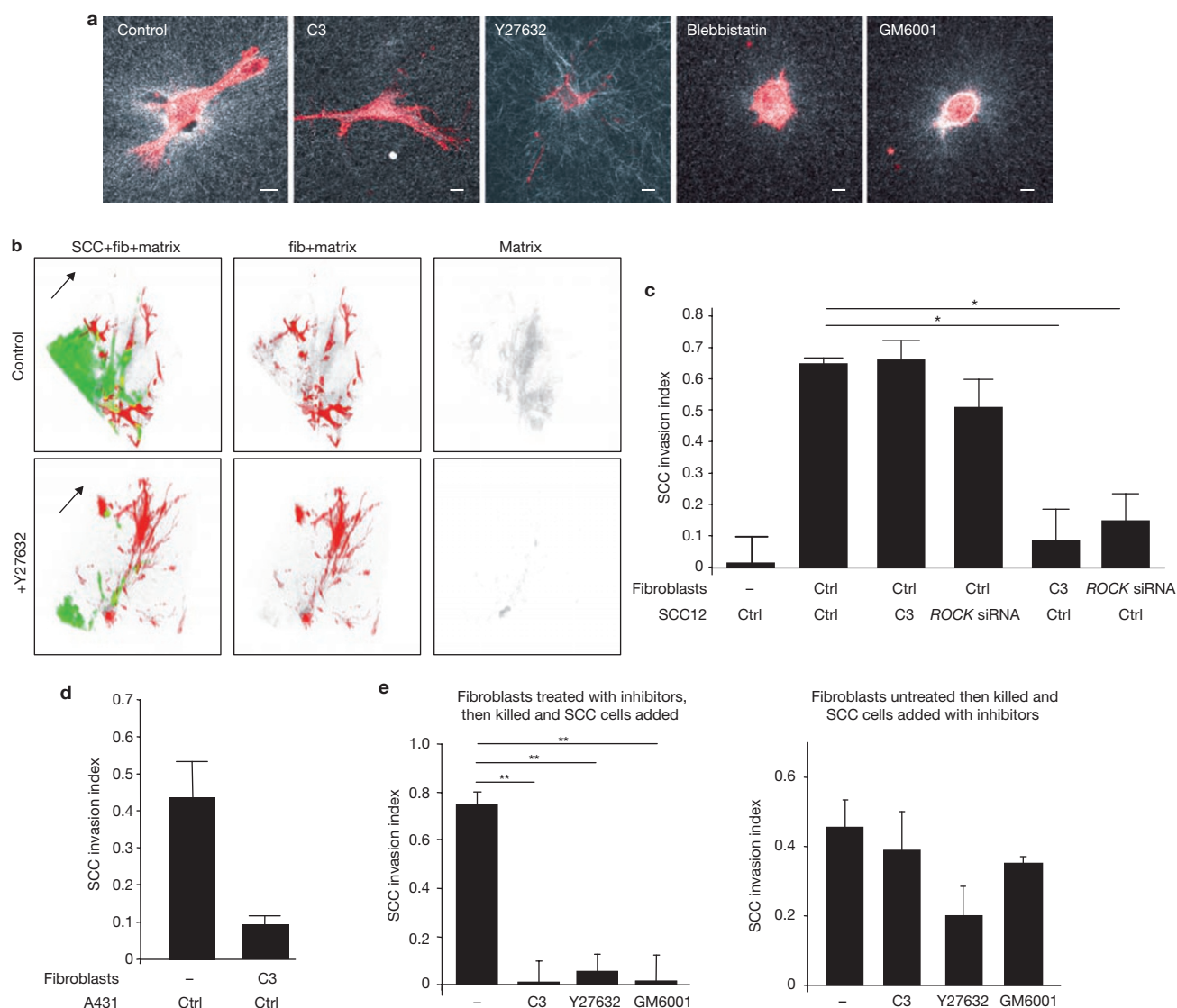


Figure 3 Rho–ROCK function is required only in stromal fibroblasts for collective SCC invasion. **(a)** Matrix remodelling and track generation by control or inhibitor-treated fibroblasts is shown by reflectance (white) and F-actin (red) imaging. The concentration of matrix and appearance of holes around control cells is indicative of matrix remodelling. **(b)** Representative images of SCC12 cells transfected with GFP–CAAX (green) and HN–CAFs transfected with mRFP–CAAX (red) invading into naïve matrix (grey in reflectance imaging) in the absence or presence of the ROCK inhibitor Y27632. The arrow indicates direction of invasion (volume rendered, approximately $200 \times 200 \times 40 \mu\text{m}$). **(c)** Quantification of SCC12-cell invasion in an organotypic assay in the absence of stromal fibroblasts, in the presence of stromal fibroblasts, with Rho function inhibited (Tat–C3 toxin) or with ROCK function inhibited (by siRNA) in either stromal fibroblasts or

SCC12 cells ($n = 3$). The single asterisk indicates $P < 0.05$. **(d)** Quantification of A431 cell invasion in an organotypic system in the presence of control stromal fibroblasts or those with Rho function inhibited by Tat–C3 toxin ($n = 2$). **(e)** Quantification of SCC12 cell invasion in matrix that previously contained stromal fibroblasts is shown on the left. Rho, ROCK or MMP function inhibited by Tat–C3 toxin, Y27632 or GM6001, respectively, in fibroblasts before their removal and addition of SCC cells ($n = 2$). Quantification of SCC12 cell invasion in matrix that previously contained stromal fibroblasts without any inhibitors. Rho, ROCK or MMP function was inhibited by Tat–C3 toxin, Y27632 or GM6001, respectively, after the fibroblasts had been removed when only SCC cells were present. The double asterisk indicates $P < 0.01$. The error bars indicate s.d. The scale bars represent $10 \mu\text{m}$ in **a**.

the matrix. In support of this hypothesis, 1×10^5 fibroblasts mixed directly with SCC cells were more effective at increasing invasion than 5×10^5 fibroblasts mixed throughout the matrix (Fig. 1e). Close examination of these sections revealed that the cell at the front of the invading chain often had an elongated morphology characteristic of a fibroblast, suggesting that fibroblasts may act as the lead cell in the collective invasion. This was directly examined by labelling fibroblasts with monomeric RFP–CAAX (mRFP–CAAX) and SCC

cells with GFP–CAAX before mixing them and imaging their invasion. Figure 1f shows an invading chain of SCC cells and fibroblasts — the leading cell was always a fibroblast (labelled in red) with SCC cells following (labelled in green; see Supplementary Information, Movie 3). Detailed analysis of >40 invading chains revealed that the leading cell was always a fibroblast and remained at the front, but that other following SCC cells often switched order in the chain (see Supplementary Information, Fig. S2).

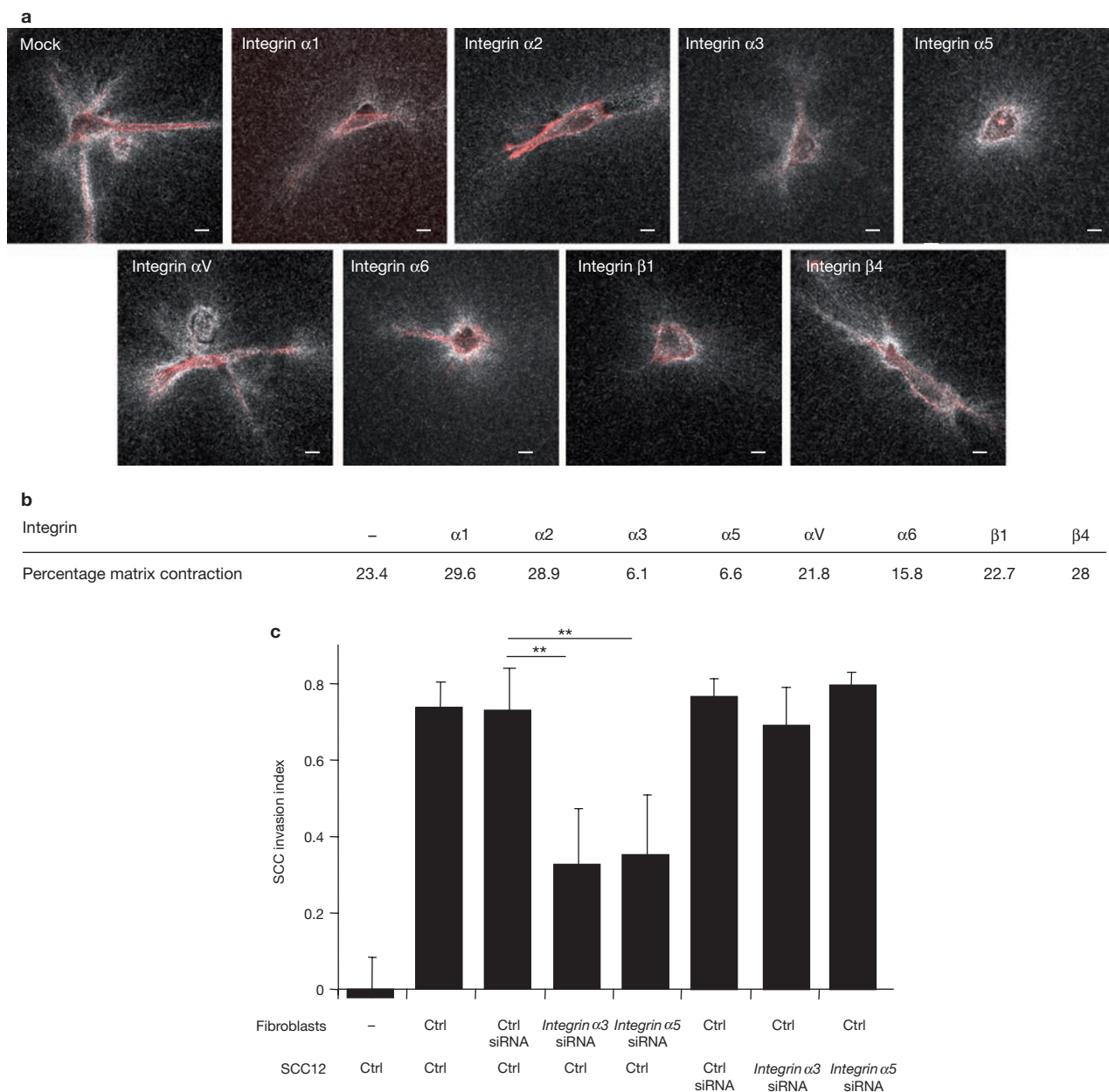


Figure 4 Integrins $\alpha 3$ and $\alpha 5$ are required in fibroblasts for collective SCC invasion. **(a)** Matrix remodelling and track generation by control or siRNA-treated fibroblasts is shown by reflectance (white) and F-actin (red) imaging. The concentration of matrix and appearance of holes around control cells is indicative of matrix remodelling. **(b)** The effect of siRNA depletion of integrins on force-mediated matrix remodelling is shown as

percentage of contraction of the gel. **(c)** Quantification of SCC12 cell invasion in an organotypic assay in the absence of stromal fibroblasts, in the presence of stromal fibroblasts and with *integrin $\alpha 3$* or *$\alpha 5$* inhibited by siRNA in either stromal fibroblasts or SCC12 cells ($n = 3$). The double asterisk indicates $P < 0.01$. The error bars indicate s.d. The scale bars represent 10 μm in **a**.

Tracks in the matrix made by fibroblasts are sufficient to enable cancer cell invasion

In the experiments described above, the matrix was also observed using reflectance imaging. Behind the leading fibroblast, an area of reduced reflectance within which the following SCC cells moved was consistently observed (Fig. 2a). There was also significant deformation of matrix usually associated with the leading fibroblast. Transmission electron microscopy was used to obtain more detailed information regarding the structure of the matrix and revealed holes in the matrix behind the fibroblast and SCC cells within these tracks (Fig. 2a). Fibroblasts were able to generate

these 'tracks' in the matrix in the absence of SCC cells (Fig. 2b) and more detailed analysis revealed that these tracks were also associated with the deposition of matrix components, such as fibronectin and tenascin-C (Fig. 2b). These data suggested that fibroblasts may be generating a track within the matrix that the SCC cells then use to invade. This hypothesis was directly examined in the organotypic system, by allowing fibroblasts five days to potentially generate tracks in the matrix before killing the fibroblasts with either puromycin or detergent extraction⁷, followed by extensive washing and seeding of SCC cells on top of the matrix. Strikingly, matrix that has been remodelled by fibroblasts that are subsequently

removed supports SCC invasion, whereas naïve mock-treated matrix does not (Fig. 2c, d). These data demonstrate that soluble factors are not responsible for fibroblast-stimulated invasion of SCC cells and suggest that physical conditioning of the ECM can promote the invasion of SCCs.

Invading fibroblasts generate tracks in the matrix that are characterized by thick collagen bundles around their sides and an absence of matrix in their centre (Fig. 2b). Reflectance imaging was used to investigate which molecular pathways are required for fibroblasts to remodel the matrix in this manner. Our previous studies demonstrated that ROCK-dependent force-mediated collagen remodelling is important for the invasion of amoeboid tumour cells⁸, and it is widely established that matrix metalloproteinases (MMPs) have a key role in matrix remodelling by fibroblasts⁹. Therefore, the effect of blocking these pathways on the generation of tracks in the matrix was investigated. Consistent with other studies^{10–12}, blockade of Rho (Tat C3), ROCK (Y27632) or non-muscle myosin (blebbistatin) function prevented the formation of holes in the ECM and concentration of matrix around the cells (Fig. 3a). These data suggest that Rho- and ROCK-dependent regulation of actomyosin interactions is required for matrix remodelling by fibroblasts. To obtain a more quantitative measure of force-mediated matrix remodelling, the ability of fibroblasts to contract the ECM was monitored. Control fibroblasts were shown to contract the matrix over a number of days (see Supplementary Information, Fig. S3a), typically by 16%, whereas Rho or ROCK inhibition reduced this to <5% (see Supplementary Information, Fig. S3b). In support of a role for Rho–ROCK signalling in regulating force-generation, of the many ROCK substrates that were examined, only changes in MLC phosphorylation correlated with the ability of stromal fibroblasts to generate tracks in the matrix (see Supplementary Information, Fig. S3c).

Rho–ROCK function is required in leading fibroblasts but not following cancer cells

Inhibition of MMP function using GM6001 prevented the formation of tracks in the matrix by fibroblasts and disrupted the normal elongated morphology of fibroblasts (Fig. 3a). GM6001 also prevented the invasion of fibroblasts into three-dimensional matrices. However, blockade of force-mediated matrix remodelling by Rho or ROCK inhibition did not prevent the invasion of stromal fibroblasts (see Supplementary Information, Fig. S3d). These data suggest that inhibition of ROCK in fibroblast–SCC cocultures should not prevent fibroblasts invading, but that SCC cells will not be able to follow them because the necessary tracks in the ECM will not be formed. Imaging of ROCK inhibitor (Y27632)-treated invasion assays with differentially labelled fibroblasts and SCC cells revealed that this was indeed the case (Fig. 3b). In both control and Y27632-treated samples, fibroblasts (in red) invade, but tracks in the matrix (in grey), and subsequent invasion of SCC cells, are only observed in the control sample. In this experimental setup, it is also possible that blockade of ROCK function in SCC cells may compromise their ability to invade in the presence of fibroblasts. Therefore, Rho or ROCK function was specifically and independently blocked in fibroblasts or SCC cells. By pretreating cells with the Rho inhibitor Tat-C3 (which remains active within cells for several days¹³) or short interfering RNA (siRNA) against *ROCK1* and *ROCK2* their function in fibroblasts was blocked, while leaving SCC cells unaffected or vice versa. Figure 3c shows that blockade of Rho or *ROCK1* and *ROCK2* function only in fibroblasts prevented the invasion of SCC12 cells. Interestingly, inhibition of Rho or *ROCK1*

and *ROCK2* in SCC12 cells did not affect the ability of SCC12 cells to invade in the presence of fibroblasts. Western-blot analysis confirmed that inhibition of Rho, *ROCK1* and *ROCK2* was efficient in both cell types (see Supplementary Information, Fig. S4a, b and data not shown). Rho function was also required in fibroblasts for their ability to promote the collective invasion of another SCC cell line, A431 (Fig. 3d). More detailed analysis of Rho isoforms revealed that RhoA, but not RhoB, is required for fibroblasts to promote SCC invasion (see Supplementary Information, Fig. S5a). Inhibition of Rho or ROCK also affected the deposition of fibronectin by fibroblasts (data not shown); however, interference with the deposition of fibronectin and tenascin-C using siRNA did not affect the ability of fibroblasts to promote SCC invasion (see Supplementary Information, Figs S4c and S5b). We believe that the deposition of these matrix components is not critical in this system; instead, the physical generation of tracks through the combined action of force- and protease-mediated matrix remodelling is crucial.

To confirm that the Rho and ROCK function was only required for generation of tracks within the ECM and not some other aspect of interaction between fibroblasts and SCC cells, the experiment in which the fibroblasts were allowed to remodel the ECM and then were removed before the addition of SCC cells was repeated. Similarly to the results described above, blockade of either Rho or ROCK function in the fibroblasts was sufficient to prevent the invasion of untreated SCC cells. In addition, inhibition of MMP function in the fibroblasts before the addition of SCC cells also prevented the invasion of SCC cells (Fig. 3e). Taken together, these results confirm the importance of both force-mediated and protease-mediated matrix remodelling by the leading fibroblasts in collectively invading cocultures. When fibroblasts had been allowed to remodel the matrix before being removed, then the invasion of SCC cells was no longer blocked by inhibition of MMP or Rho function, whereas a modest reduction in invasion was observed when ROCK function was inhibited using Y27632 (Fig. 3e). The effect of Y27632 may reflect an off-target effect of Y27632, or suggest that Rho-independent regulation of ROCK may contribute to the invasion of SCC cells; the lack of effect of *ROCK1* and *ROCK2* siRNA (Fig. 3c) suggests the former explanation is more likely. These data demonstrate that following SCC cells do not require MMP or Rho function to invade into matrix once it has been remodelled by fibroblasts.

Activation of Rho in stromal fibroblasts in collectively invading clinical samples

The data described above would predict that, in clinical samples: stromal fibroblasts should be associated with collagen-rich tracks and matrix deposition, and in close proximity to SCC cells; and that stromal fibroblasts should have active Rho signalling. These variables were analysed in frozen sections of numerous head and neck SCCs, and invading groups of SCC cells were shown to be very closely associated with fibroblasts (see Supplementary Information, Fig. S6a). Furthermore, extensive collagen bundling and fibronectin deposition was associated with the fibroblasts and there were areas lacking this dense matrix that contained SCC cells (note the similarity with Fig. 2a). To evaluate the activity of Rho proteins in the clinical samples, we used an affinity probe, based on the Rho-binding domain (RBD) of rhotekin, which only binds to Rho in its active conformation¹⁴. This probe, and a control probe with point mutations that abrogate binding to Rho, bound poorly to keratinocytes and fibroblasts in non-tumour tissue adjacent to tumours (see Supplementary

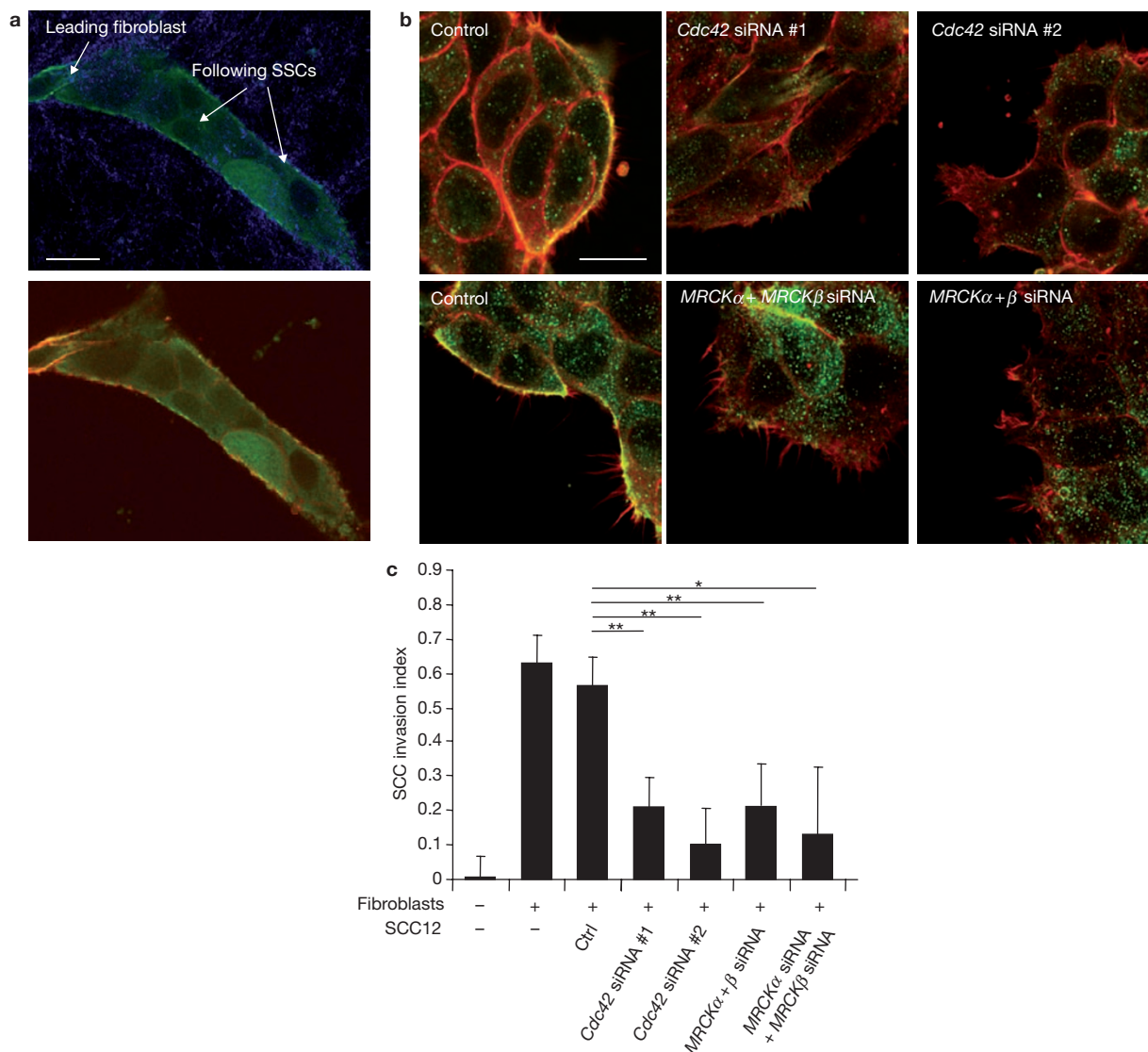


Figure 5 Cdc42-MRCK function regulates MLC and invasion in SCC cells. (a) MLC organization (green) in group of collectively invading SCC cells (matrix, blue; F-actin, red). (b) MLC phosphorylation (green) in groups of SCC cells is shown following transfection of siRNA against *Cdc42* or *MRCKα + β* (F-actin, red). (c) Quantification of SCC12-cell invasion in an organotypic system in the absence of stromal fibroblasts or in the presence

of stromal fibroblasts after transfection of siRNA against *Cdc42* or *MRCKα + β* in the SCC cells (*MRCKα + β* indicates a single siRNA that targets both *MRCKα + MRCKβ* whereas *MRCKα + MRCKβ* indicates two siRNA, one to target *MRCKα* and the other to target *MRCKβ*; $n = 5$). The single asterisk indicates $P < 0.05$ and the double asterisk $P < 0.01$. The scale bars in **a** and **b** represent 20 μm .

Information, Fig. S6b). However, significant binding of Rhotekin-RBD was observed in invading tumour tissue; notably, the probe bound to both SCC cells and also to stromal fibroblasts surrounding the tumour. In summary, analysis of clinical tumour samples revealed striking similarities to observations in our experimental system.

Integrin $\alpha 3$ and $\alpha 5$ are important in leading fibroblasts

Having established that force-mediated matrix remodelling is required in leading fibroblasts, but not in the following SCC cells, the role of the integrin family of matrix receptors was investigated. The function of integrin $\alpha 1, 2, 3, 5, V, 6$ and $\beta 1, 4$ in matrix remodelling by fibroblasts was examined using siRNA (we were unable to identify an effective siRNA against integrin $\beta 3$ in our system). Depletion of integrin $\alpha 3$ and integrin $\alpha 5$ had the most profound effect on matrix remodelling by stromal

fibroblasts, with no holes being formed in the matrix (Fig. 4a). A modest reduction in matrix remodelling was observed following transfection of integrin $\beta 1$ siRNA. However, we were unable to completely disrupt integrin $\beta 1$ function using siRNA (data not shown); therefore, the previously described role of integrin $\beta 1$ in collective invasion may be underestimated in these experiments¹⁵. Matrix remodelling was also monitored by measuring the extent of matrix contraction caused by the fibroblasts. In general, this assay corroborated the microscopic analysis of matrix remodelling, although the effect of integrin $\beta 1$ depletion was less pronounced (Fig. 4b). Specifically, this confirmed that depletion of integrins $\alpha 3$ and $\alpha 5$ prevented matrix remodelling and implicated them in the generation of contractile force by fibroblasts. Depletion of either integrin $\alpha 3$ or integrin $\alpha 5$ did not affect the ability of fibroblasts to adhere to the collagen-matrigel matrix that was used throughout this study (data not shown). Interestingly, we found

that integrin $\alpha 5$ was frequently upregulated in stromal fibroblasts in clinical SCC samples (see Supplementary Information, Fig. S6C). Based on these data, we would predict that siRNA against *integrin $\alpha 3$* or *integrin $\alpha 5$* in stromal fibroblasts would prevent the invasion of SCC cells and Figure 4C shows that siRNA-mediated depletion of either integrin $\alpha 3$ or integrin $\alpha 5$ in stromal fibroblasts, but not in SCC cells, reduced the collective invasion of SCC cells. This result was confirmed using at least two different siRNA sequences targeting *integrin $\alpha 3$* and *integrin $\alpha 5$* (see Supplementary Information, Fig. S4f, g). Thus, the requirement for integrins $\alpha 3$ or $\alpha 5$ is confined to the leading fibroblast in collectively invading cocultures. Several groups have shown that integrin $\alpha 5\beta 1$ can activate Rho signalling^{16–18}, and this result was confirmed in carcinoma-associated fibroblasts (data not shown). Integrin $\alpha 3$ was also important for force-mediated matrix remodelling, but we found no evidence that it regulated Rho activity (data not shown).

Cdc42 and MRCK function in cancer cells is required to follow fibroblasts

Having determined that integrin $\alpha 3$, integrin $\alpha 5$, MMP or Rho function are not required in SCC cells for collective invasion, we investigated which other cytoskeletal regulators might be required in SCC cells. In particular, we were interested in which pathways might be regulating MLC. Figure 5a shows that in collectively invading chains of SCC cells, MLC was clearly localized to regions of the cell cortex in contact with the ECM (Fig. 5a); therefore, various known regulators of actomyosin interactions were examined for their ability to disrupt this organization of MLC in SCC cells. Cdc42 and its effector kinases MRCK α and β have been implicated in MLC regulation in cell types that can invade in the absence of Rho function¹⁹. siRNA-mediated knockdown of either *Cdc42* or both *MRCK* isoforms clearly disrupted the localization and phosphorylation of MLC at the cortex of SCC cells (Fig. 5b and see Supplementary Information, Fig. S4d, e). Furthermore, interference with either Cdc42 or MRCK in the SCC cells blocked their ability to invade in response to stromal fibroblasts (Fig. 5c).

DISCUSSION

For many years, the hypothesis that tumour cells of epithelial origin switch to a mesenchymal phenotype to invade and metastasize has been discussed²⁰, and the tumour-promoting role of carcinoma-associated fibroblasts is well documented^{21–23}. Previous studies have focused on the production of diffusible factors by fibroblasts^{21–23}; here, we provide a new model in which physical matrix remodelling by fibroblasts enables the collective invasion of carcinoma cells that retain their epithelial phenotype (see Supplementary Information, Fig. S7). Acquisition of motile behaviour by SCC cells is not sufficient for invasion — both force-mediated and protease-mediated matrix remodelling are required. These activities are provided by stromal fibroblasts that lead collectively invading chains of cells. A combination of protease and force-mediated matrix remodelling by the leading stromal fibroblast generates a track through the matrix. Furthermore, similar tracks and extensive collagen remodelling is a common feature of invasive tumour margins and has been previously reported^{14,24–26}. These observations are not limited to SCC models, as we have observed similar results with breast carcinoma cells that have not fully converted to a mesenchymal state (410.4 and MTLn3; data not shown). Close spatial relationships between motile cell populations have also been observed in other contexts, including in breast–myofibroblast cocultures²⁴, between myeloid cells and intestinal adenocarcinoma²⁷, and in lymph nodes with leukocytes and fibroblastic reticular cells²⁸.

Blockade of integrin $\alpha 3$, $\alpha 5$ or Rho–ROCK function in stromal fibroblasts dramatically reduces the ability of fibroblasts to generate tracks in the matrix and, as a result, carcinoma cells are no longer able to follow behind fibroblasts. Consistent with this observation, we find that integrin $\alpha 5$ and Rho-GTP levels are upregulated in stromal fibroblasts in clinical samples. Matrix remodelling pathways are not required in the cells that follow the fibroblasts; consequently, SCC cells do not need to upregulate the matrix remodelling pathways that are prominent in mesenchymal cells and are able to invade without undergoing an epithelial–mesenchymal transition. Moreover, neither protease nor Rho function is required in the carcinoma cells. Instead, they use the related small G-protein Cdc42 to regulate actomyosin organization and generate the force needed to follow the tracks generated by stromal fibroblasts. This study establishes a new paradigm for collective tumour-cell invasion with different cell types performing distinct roles and requiring differing regulators of the cytoskeleton and cell–matrix adhesion. □

METHODS

Cell culture. SCC12 cells were cultured in FAD media supplemented with 10% FCS (Sera Laboratories International, Sussex, UK), 5 $\mu\text{g ml}^{-1}$ insulin (#12585-014; Invitrogen, Paisley, UK), 10 ng ml^{-1} EGF (E-9644; Sigma, Dorset, UK) and 0.5 $\mu\text{g ml}^{-1}$ hydrocortisone (H-0135; Sigma), hereafter referred to as complete media. A431 cells were grown in DMEM supplemented with 10% FCS and HN-CAF (head and neck carcinoma-associated fibroblasts) and V-CAFs (vulva carcinoma-associated fibroblasts) were cultured in DMEM supplemented with 10% FCS and insulin–transferrin–selenium (#14400-045; Invitrogen). GFP and GFP–CAAX SCC12, and mRFP–CAAX HN-CAFs were selected using G418 and FACS sorting.

Transfections, inhibitors and siRNA. SCC12 and A431 cells were cultured in standard conditions and transfected using DharmaFECT™ 2 (#T-2002-02; Dharmacon, Perbio Science, Northumberland, UK). HN-CAFs were cultured in standard conditions and transfected using DharmaFECT™ 1 (#T-2001-03; Dharmacon). Briefly, cells were plated at 60% confluence and subjected to transfection the following day using 100 nM final concentration of siRNA. siRNA was purchased from Dharmacon and sequences are listed in the Supplementary Information, Table S1. TAT-C3 toxin was added to the media 2 days before plating cells on gel¹³. Y27632 (#1254; Tocris Bioscience, Bristol, UK), H1152 (#555550; Calbiochem, UK), GM6001 (#364205; Calbiochem, Nottingham, UK) and blebbistatin (#203391; Calbiochem) were used at 10 μM , 10 μM , 5 μM and 12.5 μM , respectively.

Invasion assays. Organotypic culture system was set up as previously described^{5,29}. Briefly, 5×10^5 HN-CAFs, unless stated otherwise, were embedded in a mixture of collagen I (#354249; BD Biosciences, Oxford, UK) and Matrigel (#354234; BD Biosciences) yielding a final collagen concentration of approximately 4.6 mg ml^{-1} and a final Matrigel concentration of approximately 2.2 mg ml^{-1} . After the gel was set at 37 °C for 1 h, 5×10^5 SCC12 cells were plated on top in complete media. The gel was then mounted on a metal bridge and fed from underneath with complete media that was changed daily. After 7 days, the cultures were fixed using 4% paraformaldehyde plus 0.25% glutaraldehyde in PBS and processed by standard methods for haematoxylin and eosin (H&E) staining. For assays involving the removal of HN-CAFs, after 5 days the gels were incubated in complete media plus puromycin (5 $\mu\text{g ml}^{-1}$) for 48 h, or 1 h in extraction buffer (0.5% Triton-X100, 20 mM NH_4OH in PBS)^{7,30} to kill the fibroblasts and then washed three times with complete media (>30 min per wash). 5×10^5 SCC12 cells were then plated on top and the assays proceeded as usual. The invasion index was calculated by measuring the total area over which SCC cells had dispersed (including invading and non-invading cells) and the area of non-invading SCC cells. The value shown is the average $1 - (\text{non-invading area}/\text{total area})$ of at least ten measurements from two or more independent experiments and the error bars represent the s.d. For spheroid invasion assays, an equal ratio of SCC12 cells and HN-CAFs (25×10^3 each) were mixed in a 50 μl drop in the centre of a MatTek 35 mm dish and inverted overnight. After 24 h, the cells formed into a dense aggregate, the media was removed, the aggregate of cells was embedded in collagenI–Matrigel gel (described above) and maintained in complete media.

Invasion was imaged using confocal microscopy after 3 days of invasion. Inverted invasion assays into collagen I–Matrigel gels were used to measure the invasion of fibroblasts in the absence of SCC cells and performed, as previously described³¹, with 25×10^3 cells plated on each transwell.

Antibodies. MRCK α and MRCK β antibodies were a gift from C. Marshall (ICR, London, UK)¹⁹. Anti-fibronectin and TRITC–phalloidin were purchased from Sigma (F-3648 and P-1951, respectively). Anti-pT18S19–MLC2, anti-pEzrin–Radixin–Moesin and anti-pS3–cofilin were purchased from Cell Signaling Technology (#3674, #3141 and #3311, respectively;). Anti-Cdc42, RhoA, vimentin and anti-integrin $\alpha 5$ were purchased from Santa Cruz (sc-8401, sc-418, sc-6260 and sc-10729, respectively; Autogen Bioclear, Wiltshire, UK). Anti-ROCK1 was purchased from Chemicon (AB3885; Millipore, Warford, UK). Anti-tenascin was purchased from Abcam (ab6393; Cambridge, UK). Anti-E-cadherin (HECD-1) and integrin $\alpha 3$ (F35 177-1) were produced by Cancer Research UK antibody service (London UK).

Imaging. All imaging (except haematoxylin and eosin) was performed using inverted Zeiss LSM 510 confocal microscopes with samples contained in glass-bottomed MatTek dishes. For time-lapse imaging of organotypic cultures, a small piece of culture was cut out and embedded in collagen I–Matrigel in a glass-bottomed MatTek dish. Quantification of cell motility and three-dimensional reconstitution of invasion was performed using Volocity software, and statistical analysis using Prism software. GFP was excited using a 488 nm laser, or in cases where the area of interest was deep within the reconstituted tissue, with an 870 nm pulsed Ti–Sapphire laser. mRFP was excited with a 543 nm laser and the collagen signal was imaged using reflectance microscopy or second harmonic generation in response to an 870 nm pulsed Ti–Sapphire laser. FITC, TRITC and Cy5 fluorophores were imaged using standard lasers and filter sets. Images of H&E sections were captured using a Nikon TE2000-S equipped with a Nikon DS-5M camera. For transmission electron microscopy, samples were fixed in Sorenson's buffer with 4% paraformaldehyde and 2.5% glutaraldehyde, and processed by standard methods.

Collagen-remodelling assay. HN-CAFs (2×10^4) were embedded in collagen I–Matrigel gel. After 48 h, gels were fixed in 4% paraformaldehyde plus 0.25% glutaraldehyde before immuno- and phalloidin staining at 4 °C overnight. Collagen was visualized by reflectance microscopy using 543 nm laser excitation. To assess force-mediated matrix remodelling, 5×10^5 HN-CAFs were embedded in matrix exactly as for the organotypic assays. After 4 days, the gels were photographed, the relative diameter of the well and the gel were measured using ImageJ, and the percentage contraction was calculated using the formula $100 \times (\text{well diameter} - \text{gel diameter}) / \text{well diameter}$.

Histochemical analysis. Frozen sections head and neck SCC samples were fixed using 4% PFA in PBS for 15 min at room temperature, washed with PBS and permeabilized with 0.2% Triton X-100 in PBS for 15 min. Sections were blocked with 5% FBS in PBS before incubation with GFP (0.05 $\mu\text{g ml}^{-1}$), RBD–GFP (0.05 $\mu\text{g ml}^{-1}$)^{14,32} and RBD(AAA)–GFP (0.05 $\mu\text{g ml}^{-1}$). Fibronectin and vimentin antibodies were used at 1 $\mu\text{g ml}^{-1}$. Collagen was visualized using second harmonic generation.

Note: Supplementary Information is available on the Nature Cell Biology website.

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AUTHOR CONTRIBUTIONS

C.G. generated the majority of the data, S.H. generated data in figs 3, 4 and S1, S3 and S4. C.H.C. generated the data in fig. S6. R.G. provided reagents. J.F.M. provided technical assistance. K.H. provided clinical material for fig. S6 and carcinoma-associated fibroblasts. E.S. generated data in figs 3 and 5 and provided intellectual input.

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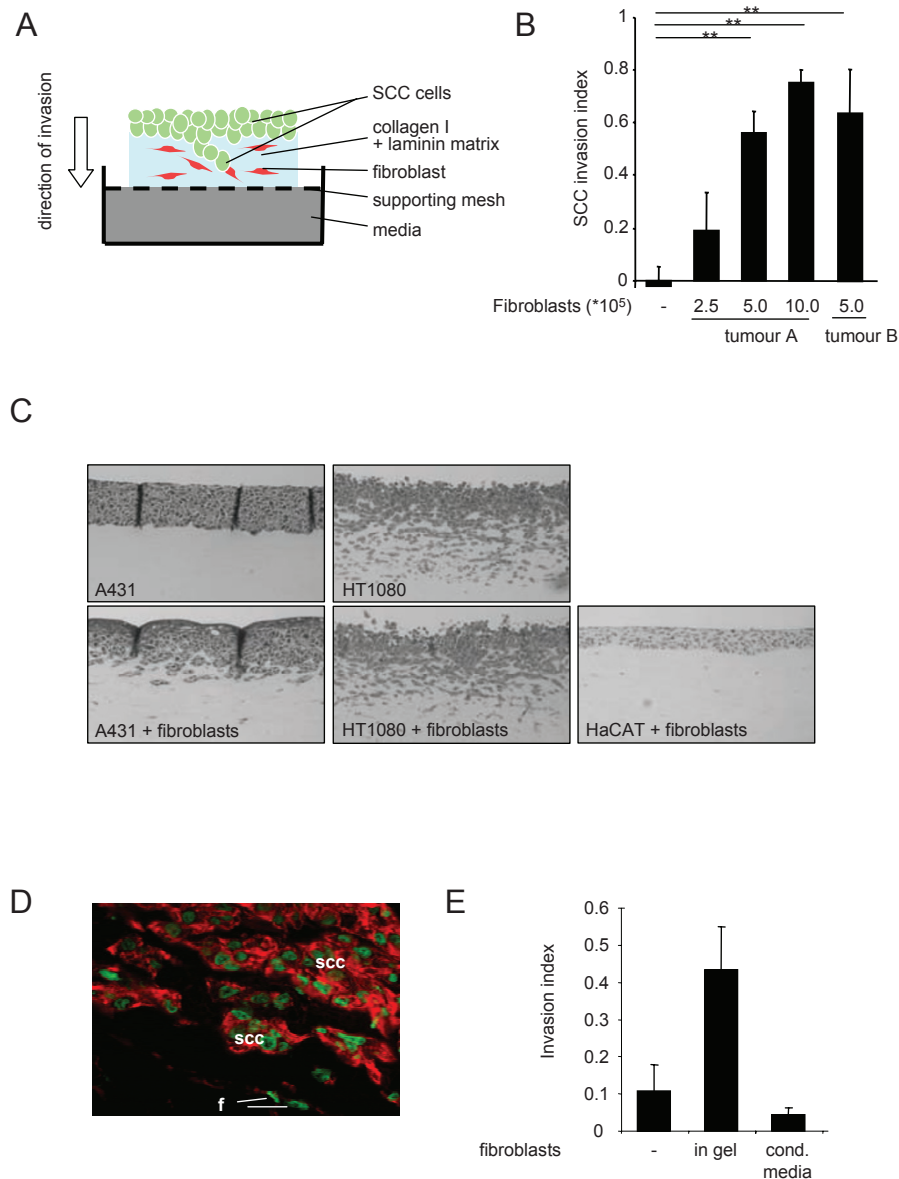


Figure S1 Organotypic culture model for invasion of carcinoma cells with epithelial characteristics. **A)** Schematic representation of organotypic culture system for studying squamous cell carcinoma invasion. **B)** Quantification of SCC12 invasion in an organotypic system in the absence, presence of varying numbers of stromal fibroblasts isolated from two different Squamous Cell Carcinomas (SCC's): Tumour A – head and neck SCC, Tumour B – vulval SCC (average of two experiments - * indicates $p < 0.05$, ** indicates $p < 0.01$).

C) Representative images of invasion of A431 (SCC cells), HaCAT (untransformed keratinocytes) and HT1080 (fibrosarcoma cells) in the presence or absence of stromal fibroblasts. **D)** E-cadherin (red) and DAPI (green) staining of collectively invading SCC cells, 'f' indicates fibroblasts 'scc' indicates squamous cell carcinoma cells. **E)** Quantification of SCC12 invasion in an organotypic system in the absence, presence of stromal fibroblasts or media conditioned by stromal fibroblasts (average of two experiments).

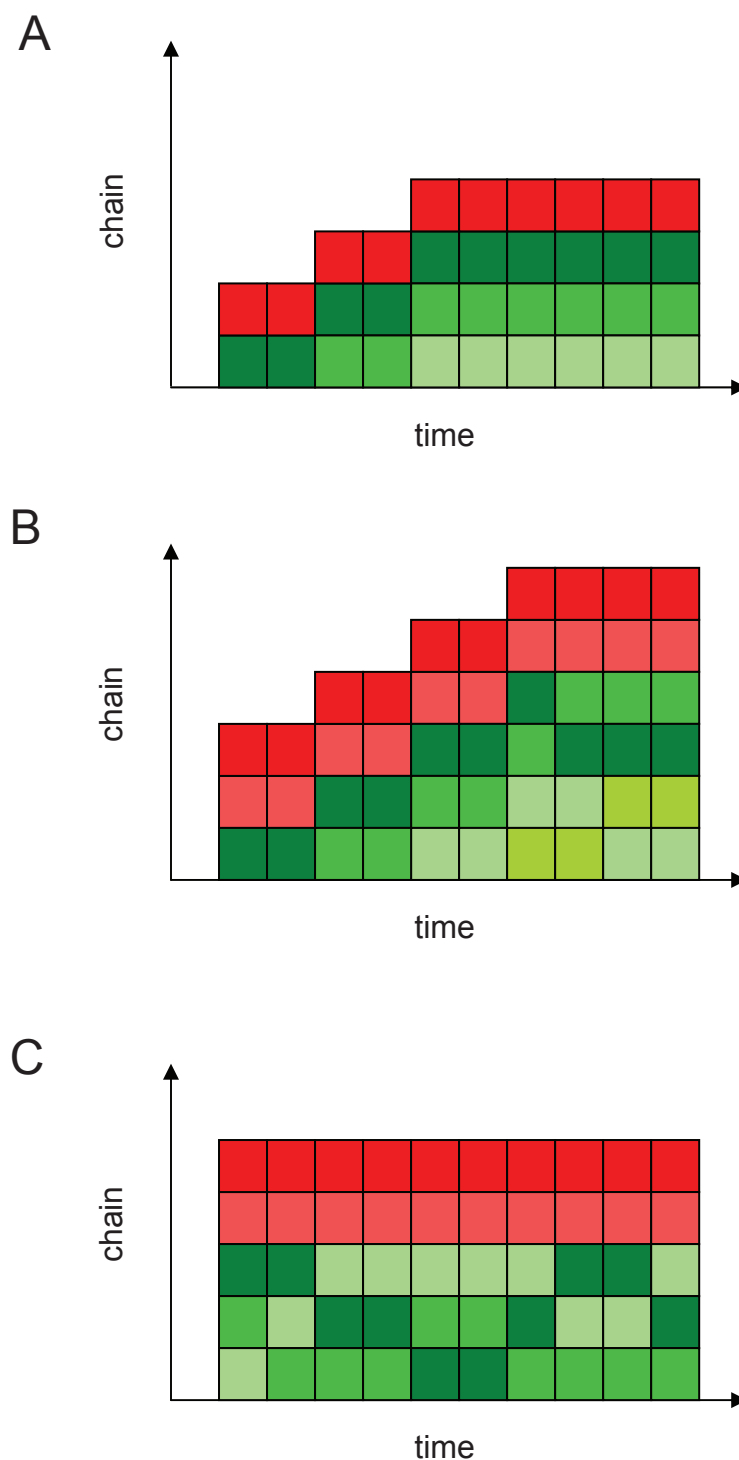


Figure S2 Spatial relationship of stromal fibroblasts and SCC cells during collective invasion. each square represents an individual cell (red shades represent stromal fibroblasts, green shades represent SCC cells), the chain length is represented by the Y axis with the leading cell uppermost, changes in cell position or number of cells in the chain over time are shown along the x axis. In example A, the chain starts with one fibroblast at the front and one

following SCC and grows by the incorporation of two SCC cells at the 'rear'. The position of cells relative to one another does not change. In example B, the chain grows by the incorporation of three SCC cells and the position of SCC cells relative to each other varies but fibroblasts always remain at the front. In example C, the number of cells in the chain does not change but the position of SCC cells relative to one another varies.

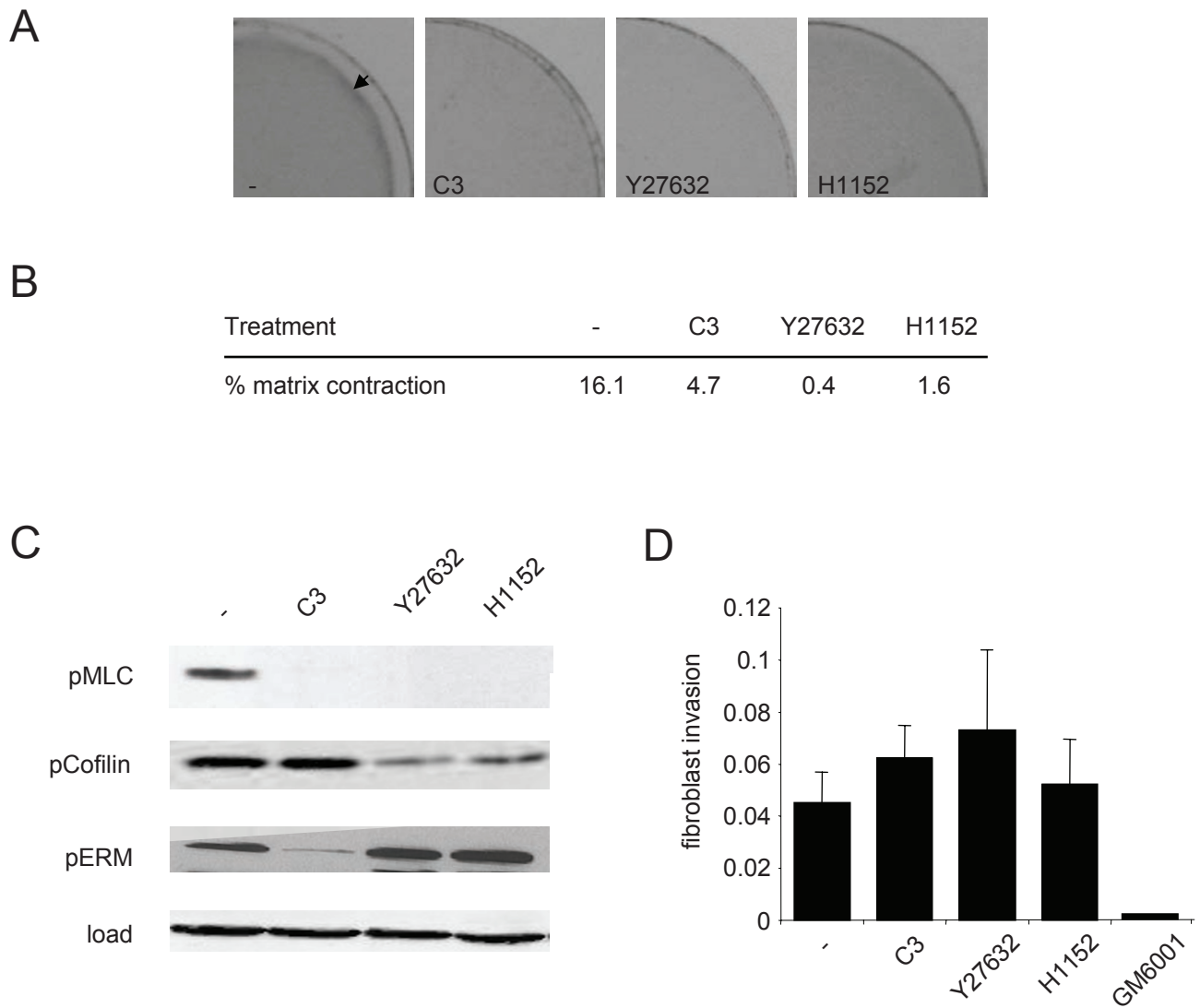
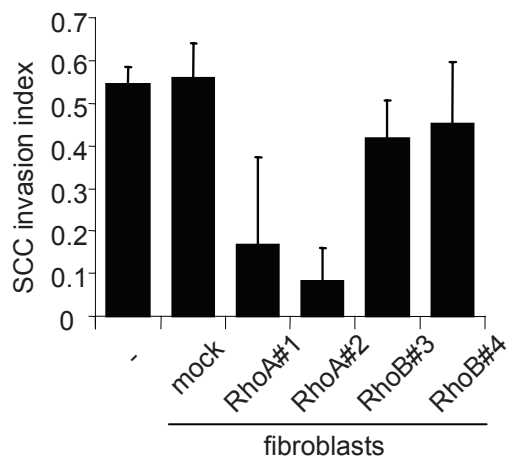


Figure S3 Role of Rho and ROCK in matrix contraction and invasion of stromal fibroblasts. A) Images of force-mediated remodeling leading to contraction of the matrix (indicated by arrow), the effect of Rho inhibition (Tat-C3) or ROCK inhibition (Y27632 or H1152) is shown. B) Table showing

% contraction of the gel. C) Effect of Tat-C3, Y27632 & H1152 on cofilin, MLC and ezrin, radixin, moesin (ERM) phosphorylation in stromal fibroblasts. D) Effect of Tat-C3, Y27632 & H1152 on stromal fibroblast invasion measured in an inverted invasion assay.

A



B

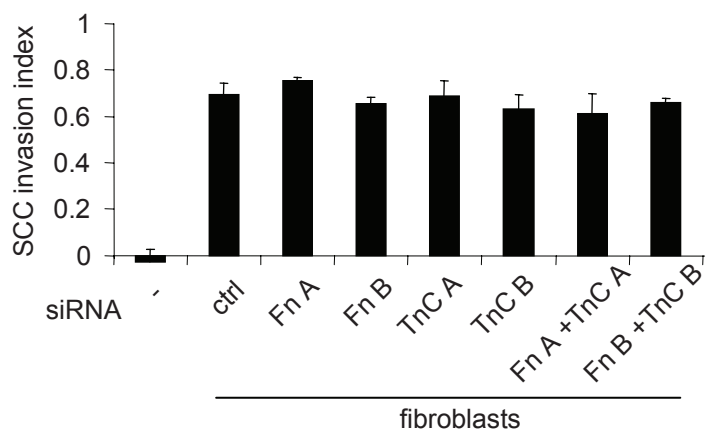


Figure S4 Role of RhoA, RhoB, Fibronectin and Tenascin expression in stromal fibroblasts. A) Quantification of SCC12 invasion in an organotypic assay in the presence of stromal fibroblasts with RhoA or RhoB depleted using siRNA in stromal fibroblasts (one of two

representative experiments). B) Quantification of SCC12 invasion in an organotypic assay in the presence of stromal fibroblasts with Fibronectin or Tenascin-C or both depleted using siRNA in stromal fibroblasts (n=2)

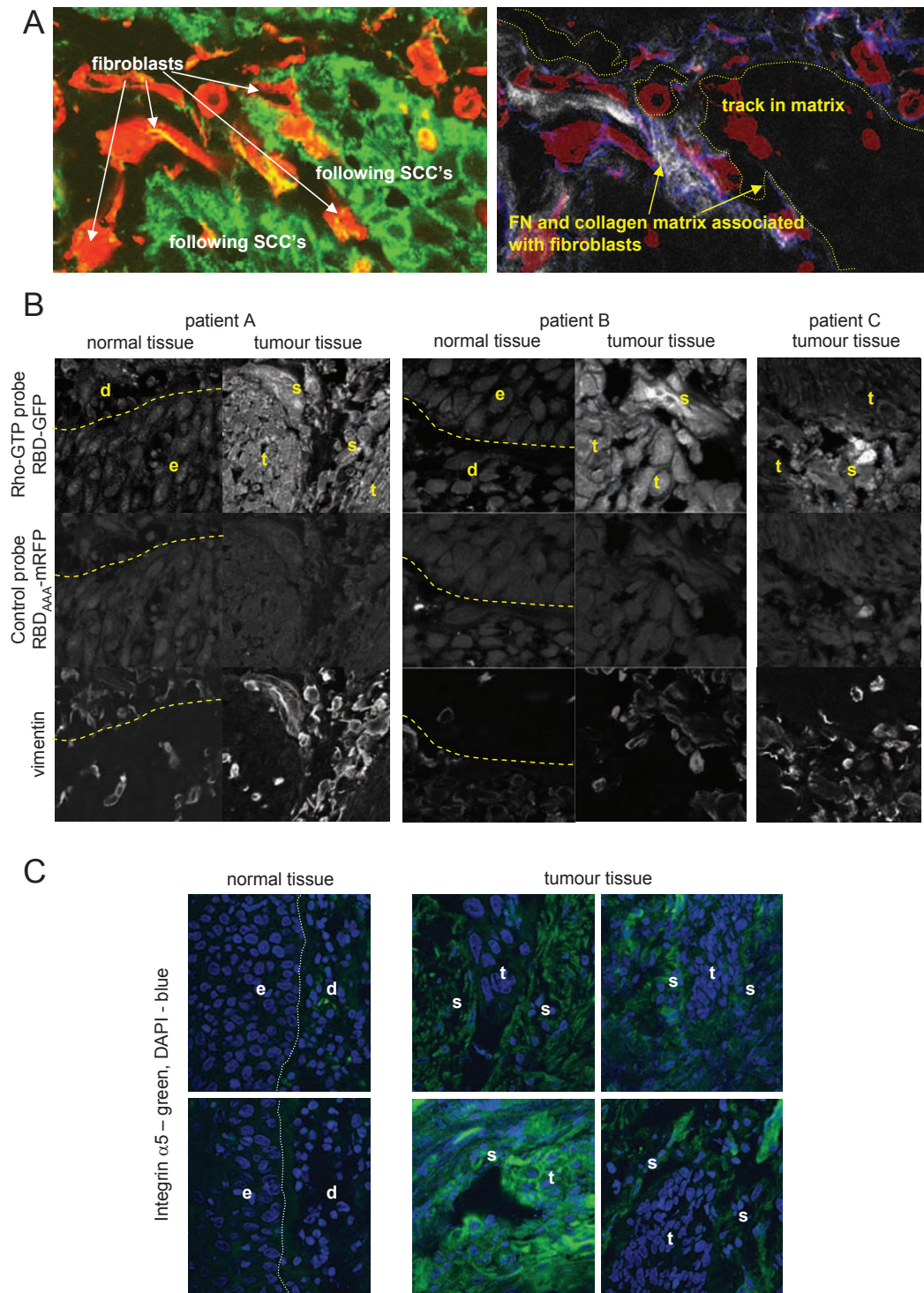
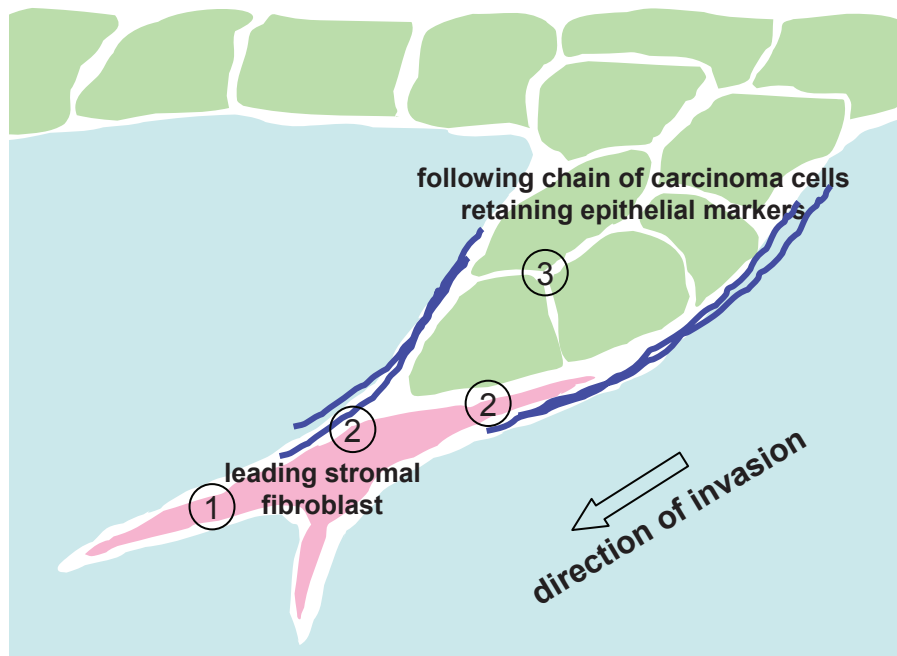


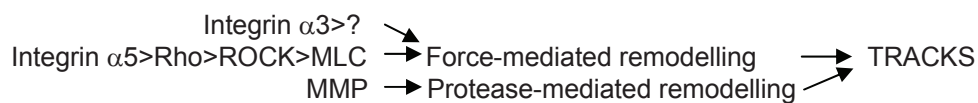
Figure S5 Analysis of stromal fibroblasts and matrix deposition in clinical samples. A) Close association of stromal fibroblasts (in red, anti-vimentin staining), collectively invading SCC cells (revealed by F-actin staining in green) and tracks in the matrix (fibrous collagen in white and fibronectin in blue) in human head and neck squamous cell carcinoma. B) Increased Rho-GTP levels in stromal fibroblasts in head and neck squamous cell carcinomas. Tumour tissue and adjacent non-tumour material was stained

with a vimentin antibody to reveal fibroblasts, the Rho-binding domain of Rhotekin fused to GFP to reveal Rho-GTP and a mutated version of the Rho-binding domain of Rhotekin fused to mRFP to control for non-specific binding. 'e' and 'd' indicate epidermis and dermis in non-tumour tissue, respectively. 't' and 's' indicate tumour and stroma, respectively. C) Increased integrin $\alpha 5$ expression in stromal fibroblasts in head and neck squamous cell carcinomas.



① Invasion of fibroblast: Rho-ROCK independent, MMP dependent

② Generation of track by fibroblast

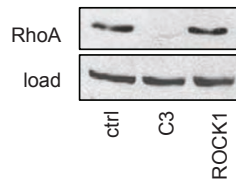


③ Carcinoma cells follow in track made by fibroblast: Cdc42-MRCK dependent

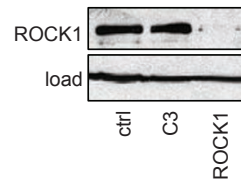
Figure S6 Model of fibroblast-led collective SCC invasion. Fibroblast is shown in pink, SCC cells in green and matrix tracks in dark blue. 1) Initially fibroblasts invade in an MMP-dependent but Rho-ROCK independent manner. 2) A track is generated in the matrix by the leading fibroblast. The generation of this depends of the function of Integrin $\alpha 5 > \text{Rho} > \text{ROCK} >$

MLC signalling, integrin $\alpha 3$ and MMP's. 3) SCC cells are unable to remodel the matrix but can utilize tracks in the matrix generated by fibroblasts and as a result follow in the tracks behind the fibroblast. SCC cells do not require the matrix redomelling activities of MMP's or Rho-ROCK to use these tracks but need Cdc42 and MRCK function.

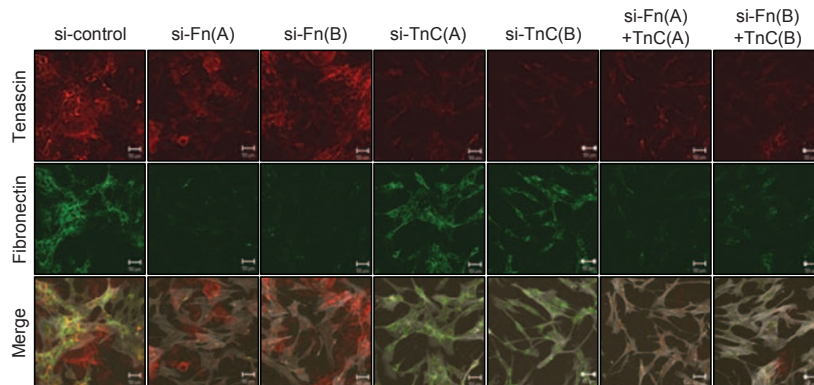
A RhoA (in HN-CAF)



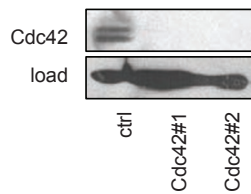
B ROCK1 (in HN-CAF)



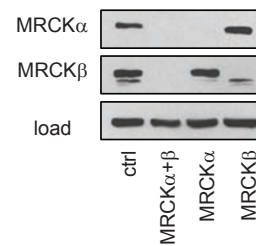
C Fibronectin and Tenascin-C (in HN-CAF)



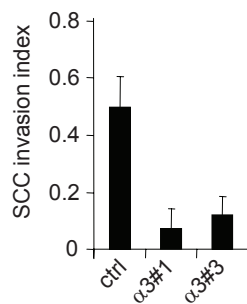
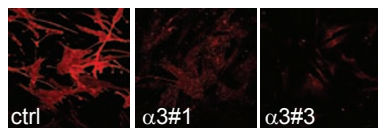
D Cdc42 (in SCC12)



E MRCK (in SCC12)



F Integrin $\alpha 3$ (in HN-CAF)



G Integrin $\alpha 5$ (in HN-CAF)

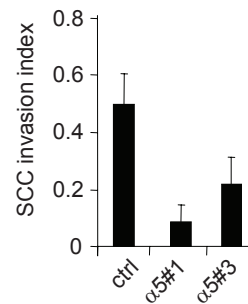
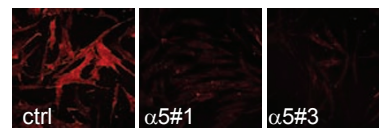
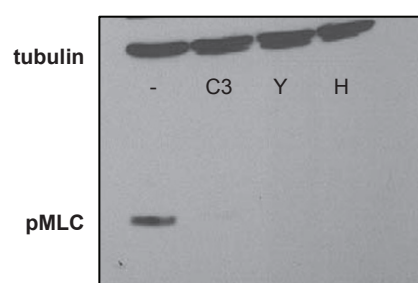
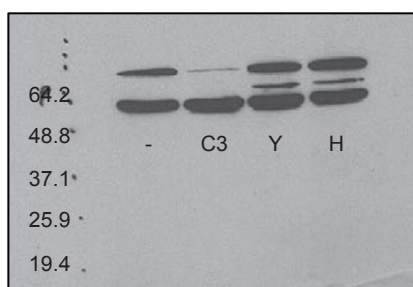
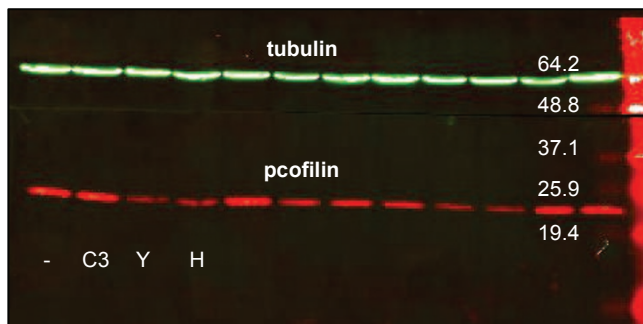


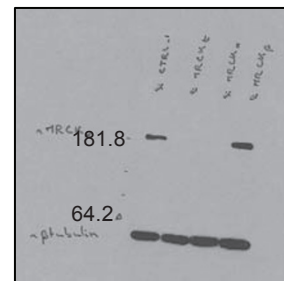
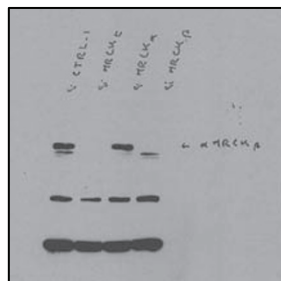
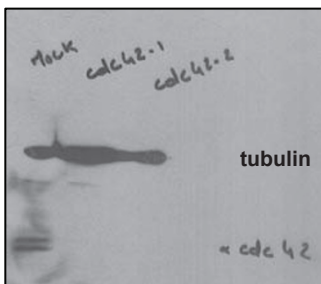
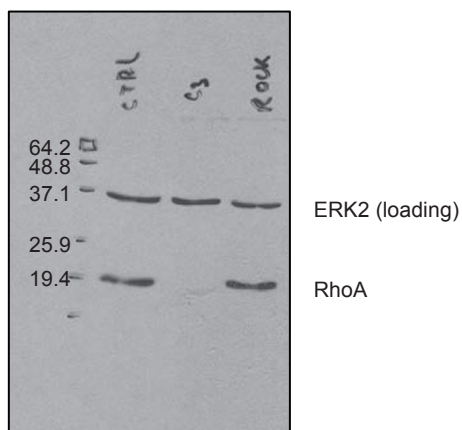
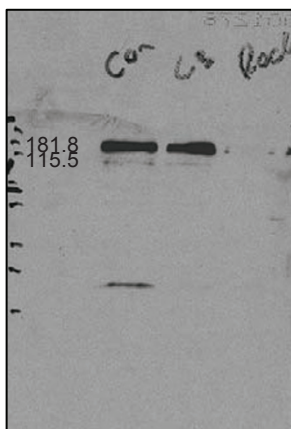
Figure S7 siRNA controls. Efficacy of RhoA, ROCK1, fibronectin, tenascin-C, Cdc42, MRCK, integrin $\alpha 3$ and integrin $\alpha 5$ knock-down by C3 or siRNA is shown by either western blot or immunofluorescence are

shown in parts A-G. Parts F and G also show that independent integrin $\alpha 3$ and $\alpha 5$ siRNA oligos are able to prevent SCC invasion when transfected into stromal fibroblasts.

Blots from Supplementary Figure 3 – Gaggioli et al.,



Blots from Supplementary Figure 6 - Gaggioli et al.,



Movie 1 SCC cells invade as collective strands. Sequential confocal sections of SCC12 and A431 cells invading in co-cultures with tumour fibroblasts (stained with phalloidin): note how all cells are incorporated into invading chains.

Movie 2 Collective invasion of SCC in an organotypic system. Time-lapse imaging of an organotypic assay with SCC cells (A431 cells in yellow) moving in a matrix containing stromal fibroblasts (in red): panel is 120x120um and spans 300 minutes.

Movie 3 Stromal fibroblasts lead the collective invasion of SCC. Time-lapse imaging of SCC12 cells (in green) and stromal fibroblasts (in red) invading into naïve matrix: panel is 250x250um and spans 240 minutes.

Supplementary Table 1: short interfering RNA sequences

RhoA#1- GAACUAUGUGGCAGAUUAUCUU
RhoA#2 – GAAAGACAUGCUUGCUCUAUUU
RhoB#3 - CAUCCAAGCCUACGACUACUU
RhoB#4 - CAACUGCUGCAAGGUGCUAUU
ROCK1 - GCCAAUGACUUACUUAGGAUU
ROCK2 - GCAAUCUGUUAAUACUCGUU
Cdc42#1- GUGGGUGCCUGAGAUAAACUUU
Cdc42#2 - AGACUCCUUUCUUGCUGUUUU
MRCKalpha and beta - CGAGAAGACUUUGAAUAAUU
MRCKalpha - AAGAAUAUCUGCUGUGUUUUU
MRCKbeta - GAAGAAUACUGAACGAAUUUU
Integrin beta1 - NNGGAACAGCAGAGAAGCUCU
Integrin beta4 smart pool
GGAAAGAGCUGCAGGUGAAUU
GAGAGCAGCUUCCAAAUCAUU
CAGAAGAUGUGGAUGAGUUUU
ACGAGAAGCUUCACACCUAUU
Integrin alpha1 smart pool
GAGGAGAGAUGGUAACUUUUU
GAAUAACGUUGGACUUUAAUU
AAAGAGAGCUUGCUAUUCAUU
GAAACAGUAUGCAUAAAUGUU
Integrin alpha2 smart pool
CAACAAGAAUGCUCAGAUUU
AAACAAGGCUGAUAAUUUGUU
GAACGGGACUUUCGCAUCAUU
CAACAUGGAUGGCUUUAAUUU
Integrin alpha3 smart pool
GCGCAAGGAGUGGGACUUUU (#1)
GGAGUGGCCCUACGAAGUCUU
UUACAGAGACUUUGACCGAUU (#3)
GUAAAUACACCGGCUACAAUU
Integrin alpha5 smart pool
UCACAUCGCUCUCAACUUCUU (#1)
ACACGUUGCUGACUCCAUUUU
GAACGAGUCAGAAUUUCGAUU (#3)
CAAACGCUCCCUCCCAUUUU
Integrin alphaV smart pool
GGAUGAAUCUGAAUUUAGAUU
CCAUGUAGAUACAAGAUUU
CGACAAAGCUGAAUGGAUUUU
CCGAAACAAUGAAGCCUUUU
Integrin alpha6 smart pool
CAACUGAAGUCACCUUUGAUU
GGAUAUGCCUCCAGGUUAAUU
CAAGACGACUCAUUAUUGAUUU
GAAAGGGAUUGUUCGUGUAUU
control#1
UAGCGACUAAACACAUCAA

Confirmation of knock-down is shown in Supplementary Figure 6